


```

> x:=t->sinh(t);
                                 $x := t \mapsto \sinh(t)$ 
(16)
=
> x(t);
                                 $\sinh(t)$ 
(17)
=
> ec1:=diff(x(t),t$4)-x(t)=0;
                                 $ec1 := 0 = 0$ 
(18)
=
> ec1;
                                 $0 = 0$ 
(19)
=
> odetest(x(t)=sinh(t),diff(x(t),t$4)-x(t)=0);
Error. (in odetest) expecting the second argument to be an ODE or a
set or list of ODEs. Received:  $0 = 0$ 
> restart;
                                restart
(20)
=
> restart;
> ec1:=diff(x(t),t$4)-x(t)=0;
                                 $ec1 := \frac{d^4}{dt^4} x(t) - x(t) = 0$ 
(21)
=
> odetest(x(t)=sinh(t),ec1);
                                 $0$ 
(22)
=
> x:=t->sinh(t);
                                 $x := t \mapsto \sinh(t)$ 
(23)
=
> odetest(x(t),ec1);
Error. (in odetest) expecting the second argument to be an ODE or a
set or list of ODEs. Received:  $0 = 0$ 
> odetest(x(t),diff(x(t),t$4)-x(t)=0);
Error. (in odetest) expecting the second argument to be an ODE or a
set or list of ODEs. Received:  $0 = 0$ 
> restart;
> x:=t->cosh(t);
                                 $x := t \mapsto \cosh(t)$ 
(24)
=
> odetest(x(t),diff(x(t),t$4)-x(t)=0);
Error. (in odetest) expecting the second argument to be an ODE or a
set or list of ODEs. Received:  $0 = 0$ 
> odetest(x(t)=cosh(t),diff(x(t),t$4)-x(t)=0);
Error. (in odetest) expecting the second argument to be an ODE or a
set or list of ODEs. Received:  $0 = 0$ 
> ec1:=diff(x(t),t$4)-x(t)=0;
                                 $ec1 := 0 = 0$ 
(25)
=
> odetest(x(t)=cosh(t),ec1);
Error. (in odetest) expecting the second argument to be an ODE or a
set or list of ODEs. Received:  $0 = 0$ 
> restart;

```

```
> odetest(x(t)=cosh(t),diff(x(t),t$4)-x(t)=0);
```

$$0 \quad (26)$$

```
> restart;
```

```
> # Exercise 2: find the general solution to each of the following
differential equations. Write each equation and its corresponding
general solution.
```

```
> # a)
```

```
> ec1:=diff(x(t),t)+t*x(t)=0;
```

$$ec1 := \frac{d}{dt} x(t) + t x(t) = 0 \quad (27)$$

```
> dsolve(ec1,x(t));
```

$$x(t) = c_1 e^{-\frac{t^2}{2}} \quad (28)$$

```
> odetest(x(t) = c__1*exp(-t^2/2), ec1);
```

$$0 \quad (29)$$

```
> restart;
```

```
> # b)
```

```
> ec2:=diff(x(t),t$2)+x(t)=0;
```

$$ec2 := \frac{d^2}{dt^2} x(t) + x(t) = 0 \quad (30)$$

```
> dsolve(ec2,x(t));
```

$$x(t) = c_1 \sin(t) + c_2 \cos(t) \quad (31)$$

```
> odetest(x(t) = c__1*sin(t) + c__2*cos(t),ec2);;
```

$$0 \quad (32)$$

```
> odetest(x(t) = c__1*sin(t) + c__2*cos(t),diff(x(t),t$2)+x(t)=0);
```

$$0 \quad (33)$$

```
> restart;
```

```
> # c)
```

```
> ec3:=4*diff(x(t),t$2)+8*diff(x(t),t)+5*x(t)=0;
```

$$ec3 := 4 \frac{d^2}{dt^2} x(t) + 8 \frac{d}{dt} x(t) + 5 x(t) = 0 \quad (34)$$

```
> dsolve(ec3,x(t));
```

$$x(t) = c_1 e^{-t} \sin\left(\frac{t}{2}\right) + c_2 e^{-t} \cos\left(\frac{t}{2}\right) \quad (35)$$

```
> odetest(x(t) = c__1*exp(-t)*sin(t/2) + c__2*exp(-t)*cos(t/2),
ec3);
```

$$0 \quad (36)$$

```
> # e)
```

```
> restart;
```

```
> ec3;
```

$$ec3 \quad (37)$$

```

> ec4:=diff(x(t),t$3)-9*diff(x(t),t$2)+15*diff(x(t),t)+25*x(t)=0;
      ec4 :=  $\frac{d^3}{dt^3} x(t) - 9 \frac{d^2}{dt^2} x(t) + 15 \frac{d}{dt} x(t) + 25 x(t) = 0$  (38)
=
> dsolve(ec4,x(t));
      x(t) =  $c_1 e^{-t} + c_2 e^{5t} + c_3 e^{5t} t$  (39)
=
> odetest(x(t) = c__1*exp(-t) + c__2*exp(5*t) + c__3*exp(5*t)*t,
ec4);
      0 (40)
=
> # d)
> ec5:=diff(x(t),t$2)-3*diff(x(t),t)+2*x(t)=0;
      ec5 :=  $\frac{d^2}{dt^2} x(t) - 3 \frac{d}{dt} x(t) + 2 x(t) = 0$  (41)
=
> dsolve(ec5,x(t));
      x(t) =  $c_1 e^t + c_2 e^{2t}$  (42)
=
> odetest(x(t) = c__1*exp(t) + c__2*exp(2*t), ec5);
      0 (43)
=
> # f)
> ec6:=diff(x(t),t$3)+diff(x(t),t$2)+diff(x(t),t)+x(t)=0;
      ec6 :=  $\frac{d^3}{dt^3} x(t) + \frac{d^2}{dt^2} x(t) + \frac{d}{dt} x(t) + x(t) = 0$  (44)
=
> dsolve(ec6, x(t));
      x(t) =  $c_1 e^{-t} + c_2 \sin(t) + c_3 \cos(t)$  (45)
=
> odetest(x(t) = c__1*exp(-t) + c__2*sin(t) + c__3*cos(t),ec6);
      0 (46)
=
> # g)
> ec7:=diff(x(t),t$3)-3*diff(x(t),t$2)+3*diff(x(t),t)-x(t)=0;
      ec7 :=  $\frac{d^3}{dt^3} x(t) - 3 \frac{d^2}{dt^2} x(t) + 3 \frac{d}{dt} x(t) - x(t) = 0$  (47)
=
> dsolve(ec7,x(t));
      x(t) =  $c_1 e^t + c_2 e^t t + c_3 e^t t^2$  (48)
=
> odetest(x(t) = c__1*exp(t) + c__2*exp(t)*t + c__3*exp(t)*t^2,
ec7);
      0 (49)
=
> # g)
> ec8:=diff(x(t),t$3)-3*diff(x(t),t$2)+3*diff(x(t),t)-x(t)=0;
      ec8 :=  $\frac{d^3}{dt^3} x(t) - 3 \frac{d^2}{dt^2} x(t) + 3 \frac{d}{dt} x(t) - x(t) = 0$  (50)
=
> dsolve(ec8,x(t));
      x(t) =  $c_1 e^t + c_2 e^t t + c_3 e^t t^2$  (51)

```

```
> odetest(x(t) = c__1*exp(t) + c__2*exp(t)*t + c__3*exp(t)*t^2,
ec8);
0 (52)
```

```
=
> # h)
> ec9:=diff(x(t),t$3)-9*diff(x(t),t$2)+15*diff(x(t),t)+25*x(t)=0;
ec9 :=  $\frac{d^3}{dt^3} x(t) - 9 \frac{d^2}{dt^2} x(t) + 15 \frac{d}{dt} x(t) + 25 x(t) = 0$  (53)
```

```
=
> dsolve(ec9,x(t));
x(t) = c1 e-t + c2 e5t + c3 e5t t (54)
```

```
=
> odetest(x(t) = c__1*exp(-t) + c__2*exp(5*t) + c__3*exp(5*t)*t,
ec9);
0 (55)
```

```
=
> # i)
> ec10:=diff(x(t),t$4)-lambda^4 * x(t)=0;
ec10 :=  $\frac{d^4}{dt^4} x(t) - \lambda^4 x(t) = 0$  (56)
```

```
=
> dsolve(ec10,x(t));
x(t) = c1 e-λt + c2 eλt + c3 sin(λ t) + c4 cos(λ t) (57)
```

```
=
> odetest(x(t)=c__1*exp(-lambda*t) + c__2*exp(lambda*t) + c__3*sin
(lambda*t) + c__4*cos(lambda*t),ec10);
0 (58)
```

```
=
> # j)
> ec11:=diff(x(t),t$4)+5*diff(x(t),t$2)+4*x(t)=0;
ec11 :=  $\frac{d^4}{dt^4} x(t) + 5 \frac{d^2}{dt^2} x(t) + 4 x(t) = 0$  (59)
```

```
=
> dsolve(ec11,x(t));
x(t) = c1 sin(t) + c2 cos(t) + c3 sin(2 t) + c4 cos(2 t) (60)
```

```
=
> odetest(x(t) = c__1*sin(t) + c__2*cos(t) + c__3*sin(2*t) + c__4*
cos(2*t),ec11);
0 (61)
```

```
=
> restart;
```

```
> # Exercise 3;
```

```
> # find solution to each ivp and then represent its graph on a
large interval; if not accurate graph, find zeros of the
function, its limit at infinity, portions of graphs on small
intervals; describe long-term behavior: exponential
increasing/decreasing; periodic with minimal period...; bounded
between... ; oscillatory around zero, with constant amplitude,
with amplitude decreasing to zero as t-> infinity, with amplitude
increasing to infinity as t-> infinity
```

```
> # a)
```

```
> ec1:=diff(x(t),t$2)+x(t)=0;
```

$$ec1 := \frac{d^2}{dt^2} x(t) + x(t) = 0 \quad (62)$$

```
> incond:=x(Pi/2)=1,D(x)(Pi/2)=-2;
```

$$incond := x\left(\frac{\pi}{2}\right) = 1, D(x)\left(\frac{\pi}{2}\right) = -2 \quad (63)$$

```
> sol:=dsolve({ec1,incond},x(t));
```

$$sol := x(t) = \sin(t) + 2 \cos(t) \quad (64)$$

```
> x:=unapply(rhs(sol),l);
```

$$x := l \mapsto \sin(t) + 2 \cdot \cos(t) \quad (65)$$

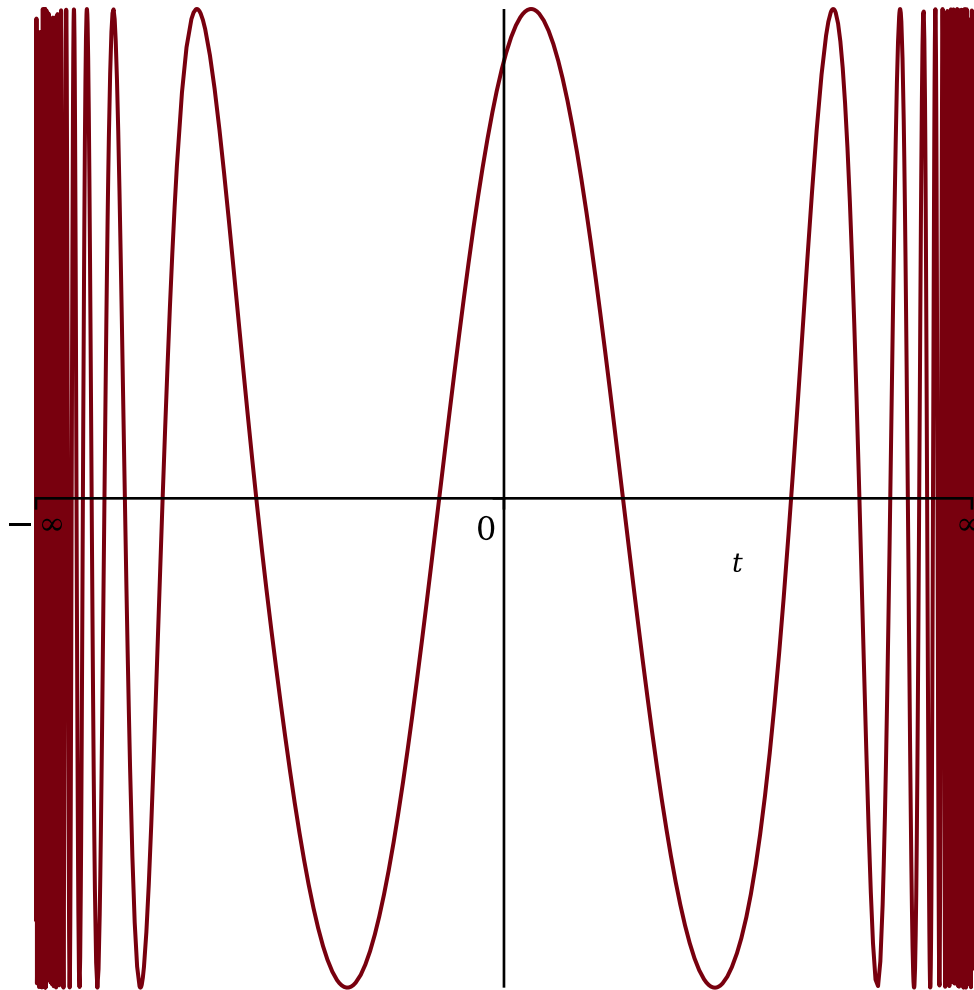
```
> x(t);
```

$$\sin(t) + 2 \cos(t) \quad (66)$$

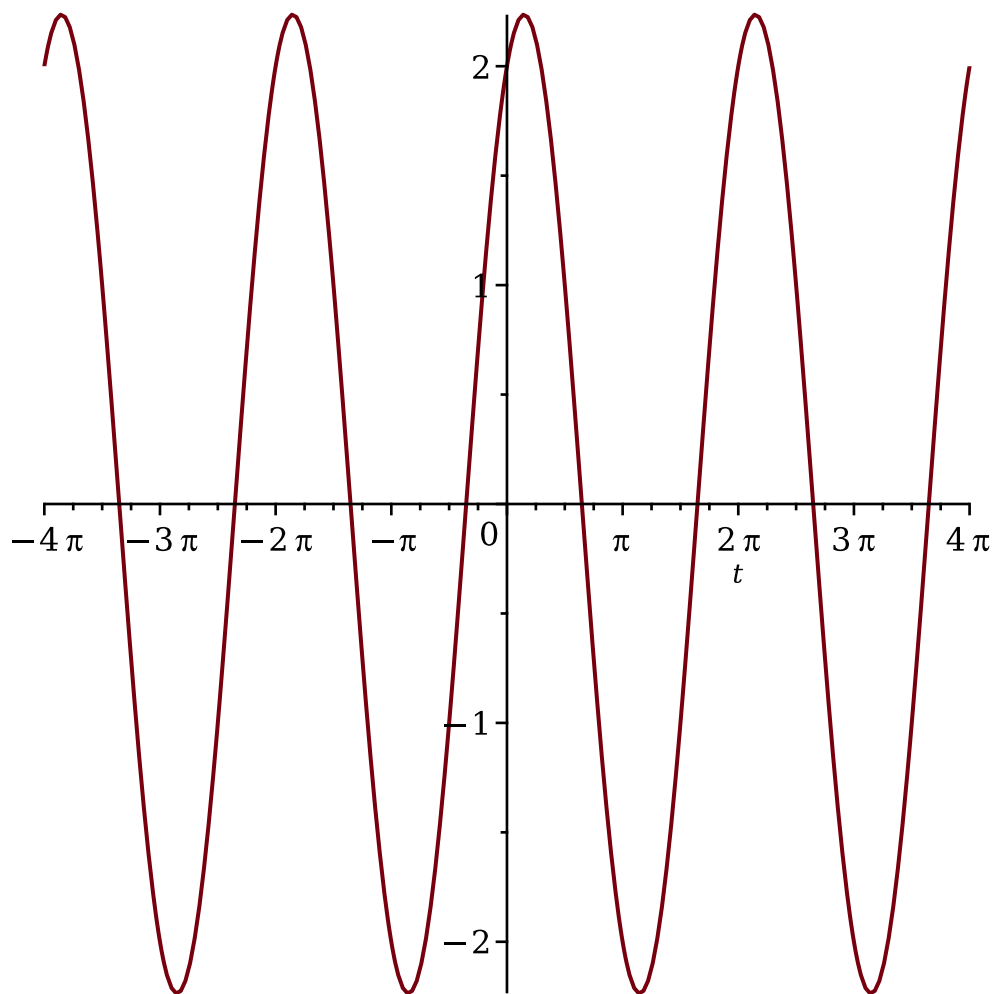
```
> limit(x(t),t=infinity);
```

$$-3.3 \quad (67)$$

```
> plot(x(t),t=-infinity..infinity);
```



```
> plot(x(t),t=-4*Pi..4*Pi);
```



```

> restart;
> # b)
> ec:=4*diff(x(t),t$2)+8*diff(x(t),t)+5*x(t)=0;

```

$$ec := 4 \frac{d^2}{dt^2} x(t) + 8 \frac{d}{dt} x(t) + 5 x(t) = 0 \quad (68)$$

```

> incond:=x(0)=0,D(x)(0)=0.5;
      incond := x(0) = 0, D(x)(0) = 0.5

```

```

> sol:=dsolve({ec,incond},x(t));
      sol := x(t) = e^{-t} sin\left(\frac{t}{2}\right)

```

```

> x:=unapply(rhs(sol),t);
      x := t ↦ e^{-t} · sin\left(\frac{t}{2}\right)

```

```

> x(t);
      e^{-t} sin\left(\frac{t}{2}\right)

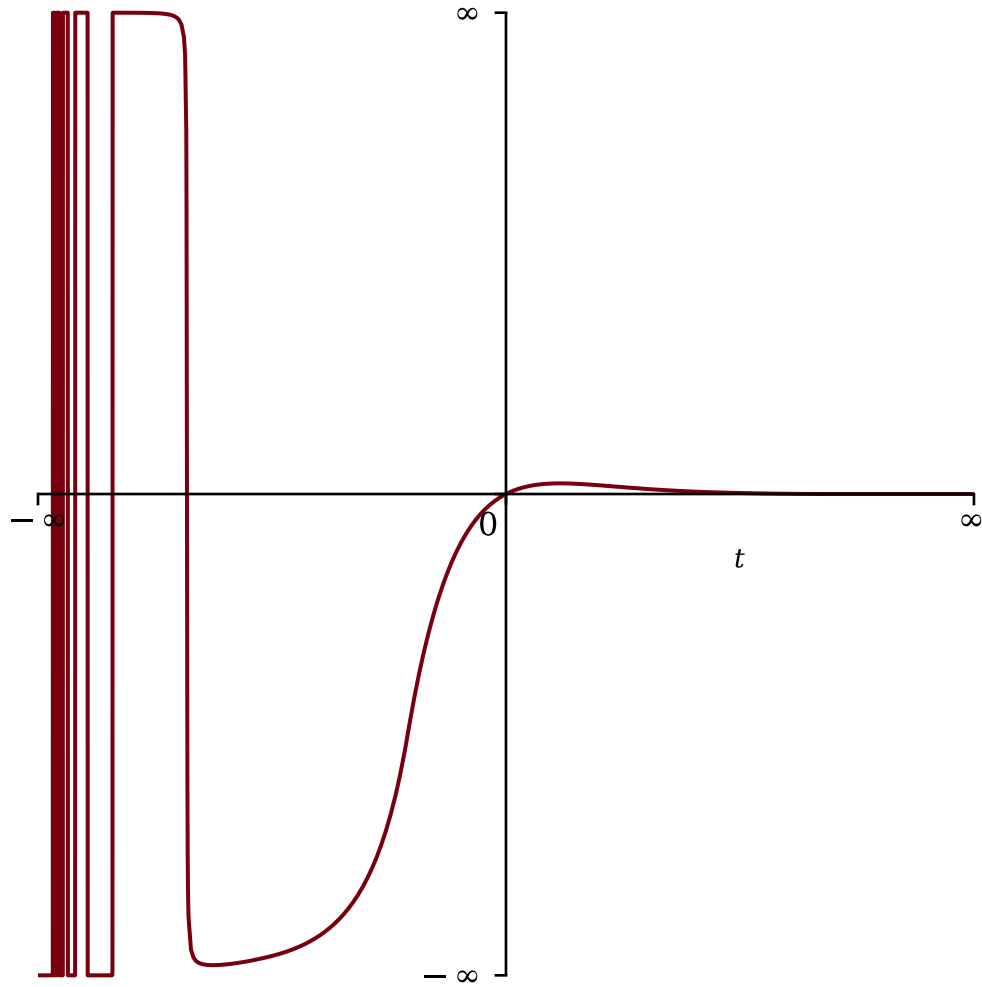
```

```
> limit(x(t),t=infinity);
```

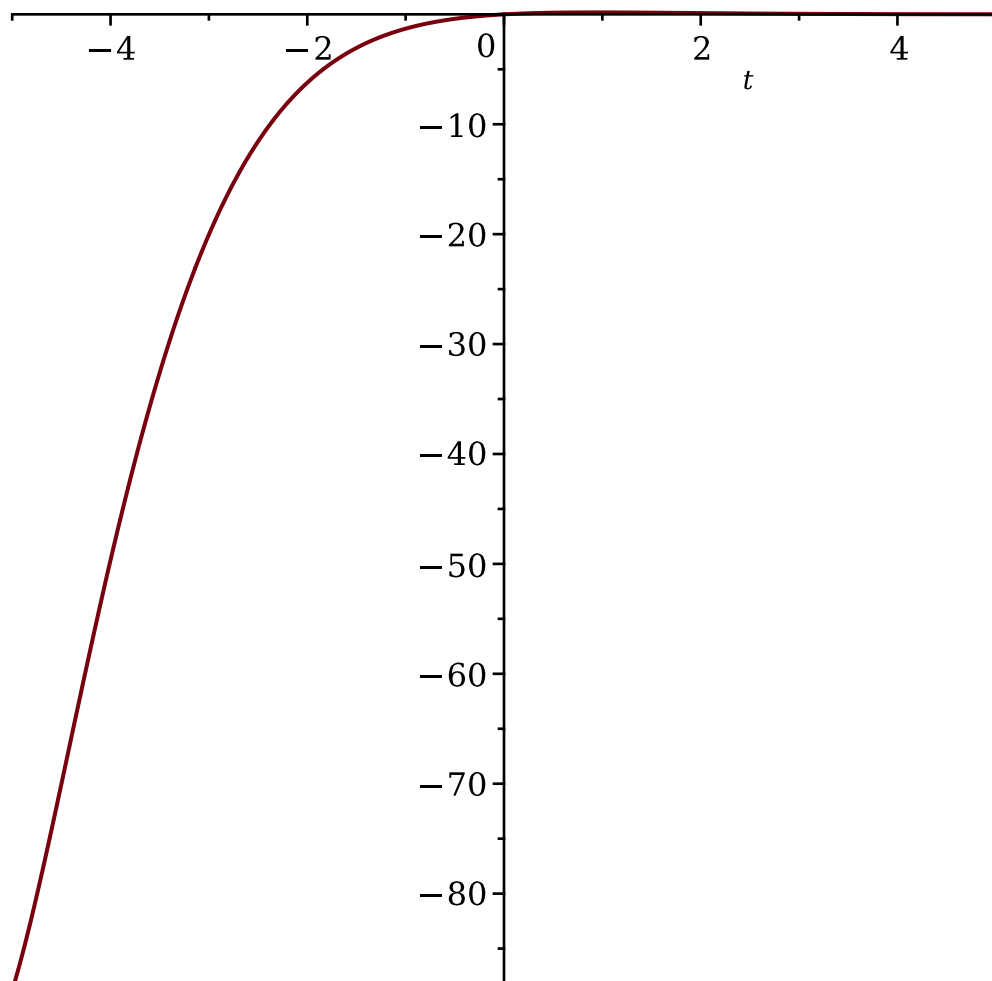
0

(73)

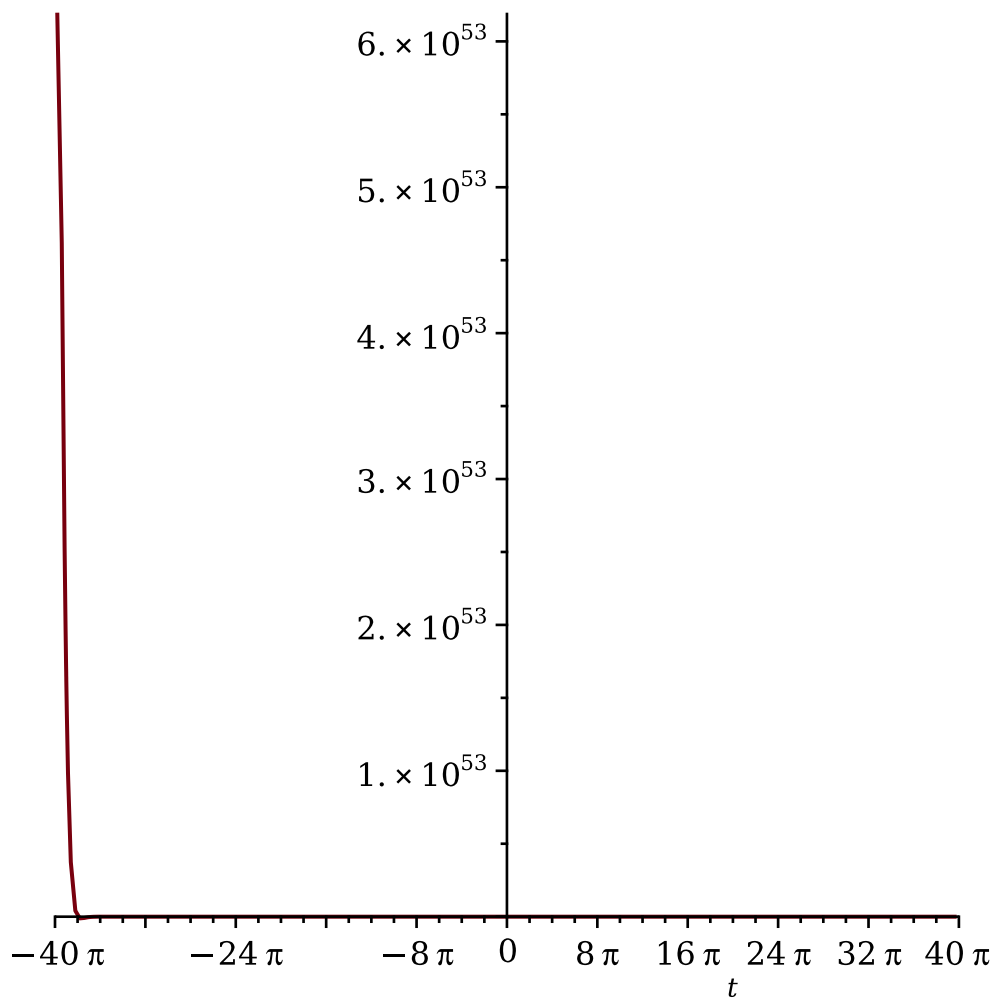
```
> plot(x(t),t=-infinity..infinity);
```



```
> plot(x(t),t=-5..5);
```

```
> plot(x(t),t=-40*Pi..40*Pi);
```



```
> restart;
```

```
> # c)
```

```
> ec:=diff(x(t),t$2)-3*diff(x(t),t)+2*x(t)=0;
```

$$ec := \frac{d^2}{dt^2} x(t) - 3 \frac{d}{dt} x(t) + 2 x(t) = 0 \quad (74)$$

```
> incond:=x(0)=2,D(x)(0)=3;
```

$$incond := x(0) = 2, D(x)(0) = 3 \quad (75)$$

```
> sol:=dsolve({ec,incond},x(t));
```

$$sol := x(t) = e^{2t} + e^t \quad (76)$$

```
> x:=unapply(rhs(sol),t);
```

$$x := t \mapsto e^{2 \cdot t} + e^t \quad (77)$$

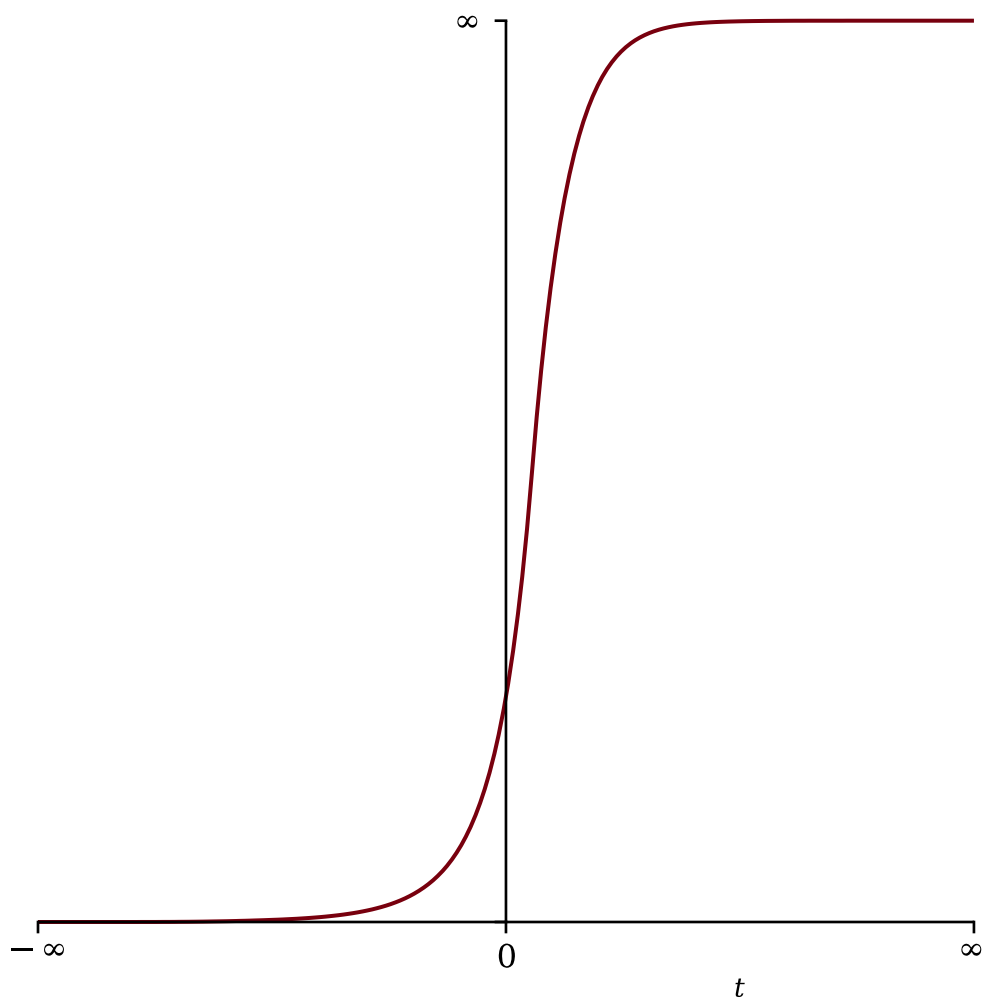
```
> x(t);
```

$$e^{2t} + e^t \quad (78)$$

```
> limit(x(t),t=infinity);
```

$$\infty \quad (79)$$

```
> plot(x(t),t=-infinity..infinity);
```



```
> # d)
> restart;
> ec:=diff(x(t),t$3)-3*diff(x(t),t$2)+3*diff(x(t),t)-x(t)=0;
      
$$ec := \frac{d^3}{dt^3} x(t) - 3 \frac{d^2}{dt^2} x(t) + 3 \frac{d}{dt} x(t) - x(t) = 0$$
 (80)
```

```
> incond:=x(0)=0,D(x)(0)=-1,(D@@2)(x)(0)=2;
      
$$incond := x(0) = 0, D(x)(0) = -1, D^{(2)}(x)(0) = 2$$
 (81)
```

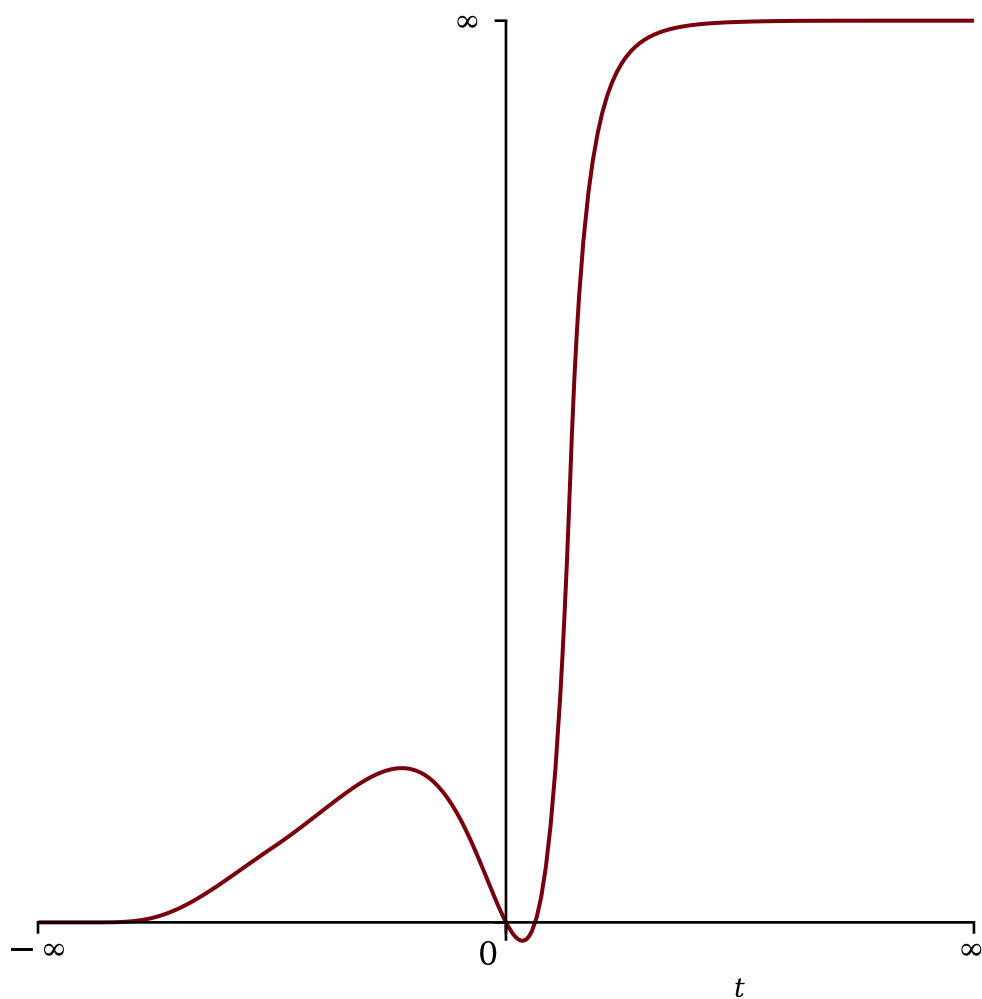
```
> sol:=dsolve({ec,incond},x(t));
      
$$sol := x(t) = e^t t (2 t - 1)$$
 (82)
```

```
> x:=unapply(rhs(sol),t);
      
$$x := t \mapsto e^t \cdot t \cdot (2 \cdot t - 1)$$
 (83)
```

```
> x(t);
      
$$e^t t (2 t - 1)$$
 (84)
```

```
> limit(x(t),t=infinity);
      
$$\infty$$
 (85)
```

```
> plot(x(t),t=-infinity..infinity);
```



```
> restart;
```

```
> # e)
```

```
> restart
```

```
> ec:=diff(x(t),t$4)+5*diff(x(t),t$2)+4*x(t)=0;
```

$$ec := \frac{d^4}{dt^4} x(t) + 5 \frac{d^2}{dt^2} x(t) + 4 x(t) = 0 \quad (86)$$

```
> incond:=x(0)=-1,D(x)(0)=0,(D@@2)(x)(0)=1,(D@@3)(x)(0)=0;
```

$$incond := x(0) = -1, D(x)(0) = 0, D^{(2)}(x)(0) = 1, D^{(3)}(x)(0) = 0 \quad (87)$$

```
> sol:=dsolve({ec,incond},x(t));
```

$$sol := x(t) = -\cos(t) \quad (88)$$

```
> x:=unapply(rhs(sol),t);
```

$$x := t \mapsto -\cos(t) \quad (89)$$

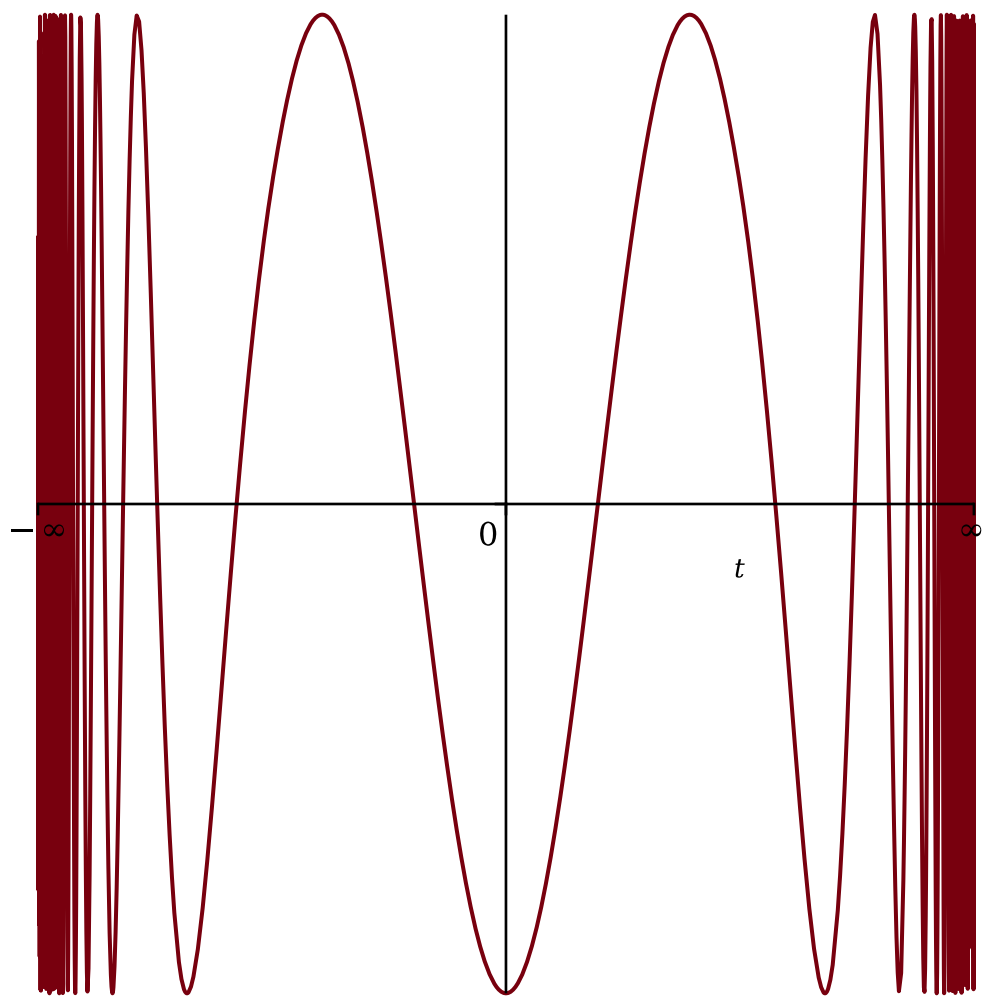
```
> x(t);
```

$$-\cos(t) \quad (90)$$

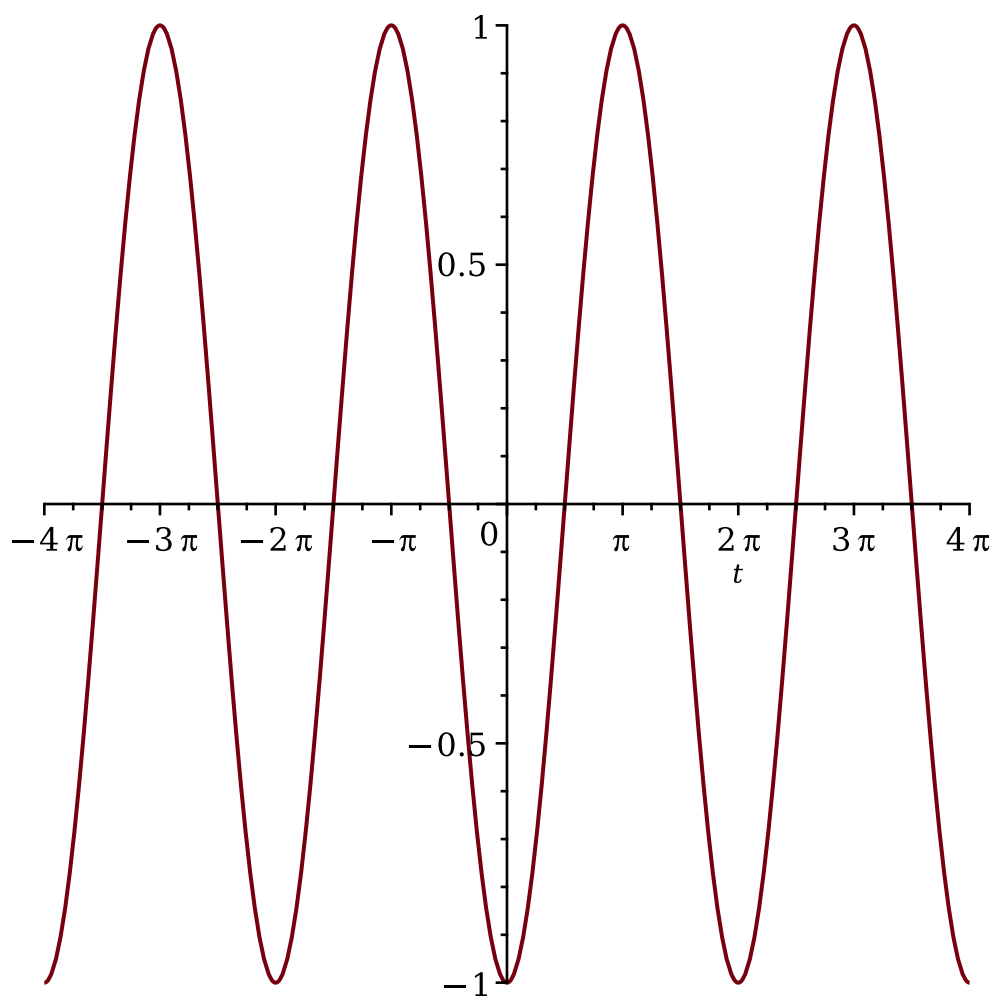
```
> limit(x(t),t=infinity);
```

$$-1..1 \quad (91)$$

```
> plot(x(t),t=-infinity..infinity);
```



```
> plot(x(t),t=-4*Pi..4*Pi);
```

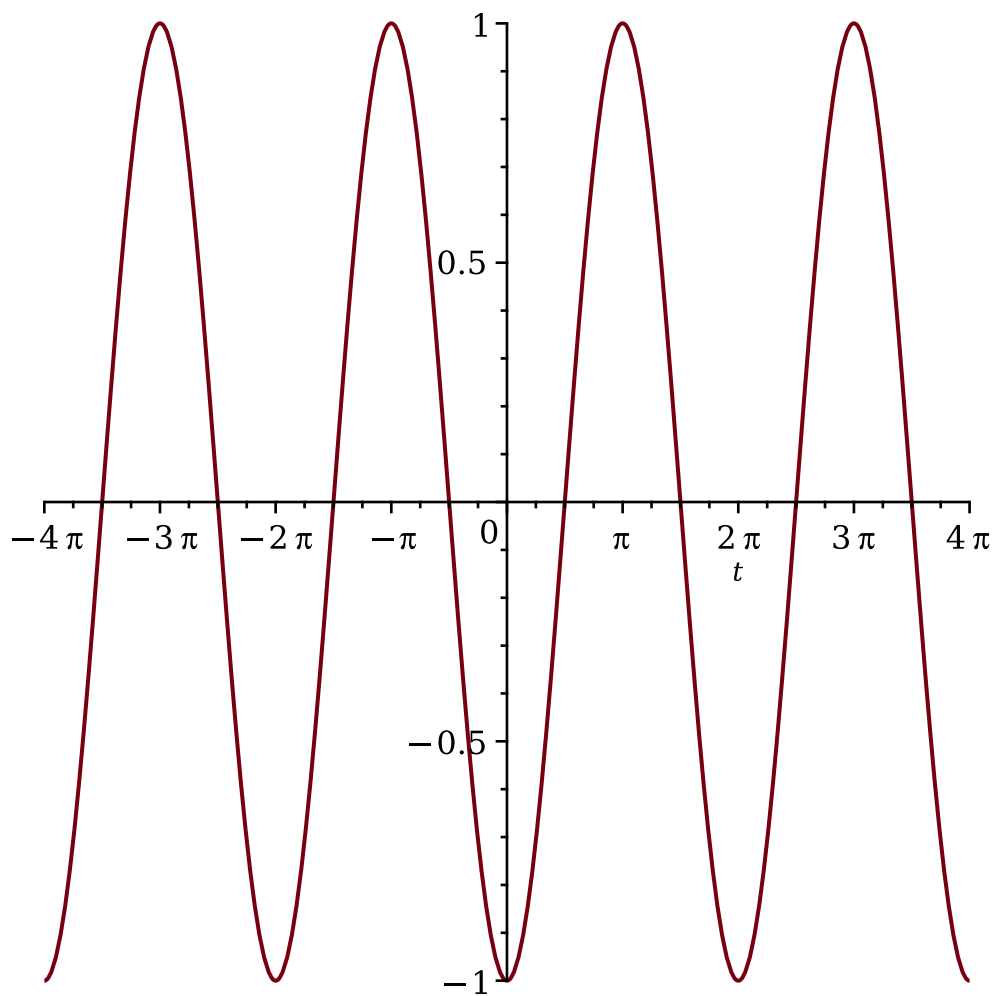


```
> simplify(rhs(sol),trig);
```

$-\cos(t)$

(92)

```
> plot(rhs(sol),t=-4*Pi..4*Pi);
```



```
> restart;
> # Exercise 4
> infolevel[dsolve]:=3
```

infolevel_{dsolve} := 3

(93)

```
> # a)
> eqa:=diff(x(t),t$2)+5*x(t)=0;
```

$$eqa := \frac{d^2}{dt^2} x(t) + 5 x(t) = 0$$

(94)

```
> dsolve(eqa,x(t));
Methods for second order ODEs:
--- Trying classification methods ---
trying a quadrature
checking if the LODE has constant coefficients
<- constant coefficients successful
```

$$x(t) = c_1 \sin(\sqrt{5} t) + c_2 \cos(\sqrt{5} t)$$

(95)

```
> # b)
> eqb:=diff(x(t),t$2)+t*x(t)=0;
```

(96)

$$eqb := \frac{d^2}{dt^2} x(t) + tx(t) = 0 \quad (96)$$

```
> dsolve(eqb,x(t));
```

Methods for second order ODEs:

--- Trying classification methods ---

trying a quadrature

checking if the LODE has constant coefficients

checking if the LODE is of Euler type

trying a symmetry of the form [xi=0, eta=F(x)]

checking if the LODE is missing 'y'

-> Trying a Liouvillian solution using Kovacic's algorithm

<- No Liouvillian solutions exists

-> Trying a solution in terms of special functions:

-> Bessel

<- Bessel successful

<- special function solution successful

$$x(t) = c_1 \text{AiryAi}(-t) + c_2 \text{AiryBi}(-t) \quad (97)$$

```
> # c
```

```
> eqc:=diff(x(t),t$2)+t^5*x(t)=0;
```

$$eqc := \frac{d^2}{dt^2} x(t) + t^5 x(t) = 0 \quad (98)$$

```
> dsolve(eqc,x(t));
```

Methods for second order ODEs:

--- Trying classification methods ---

trying a quadrature

checking if the LODE has constant coefficients

checking if the LODE is of Euler type

trying a symmetry of the form [xi=0, eta=F(x)]

checking if the LODE is missing 'y'

-> Trying a Liouvillian solution using Kovacic's algorithm

<- No Liouvillian solutions exists

-> Trying a solution in terms of special functions:

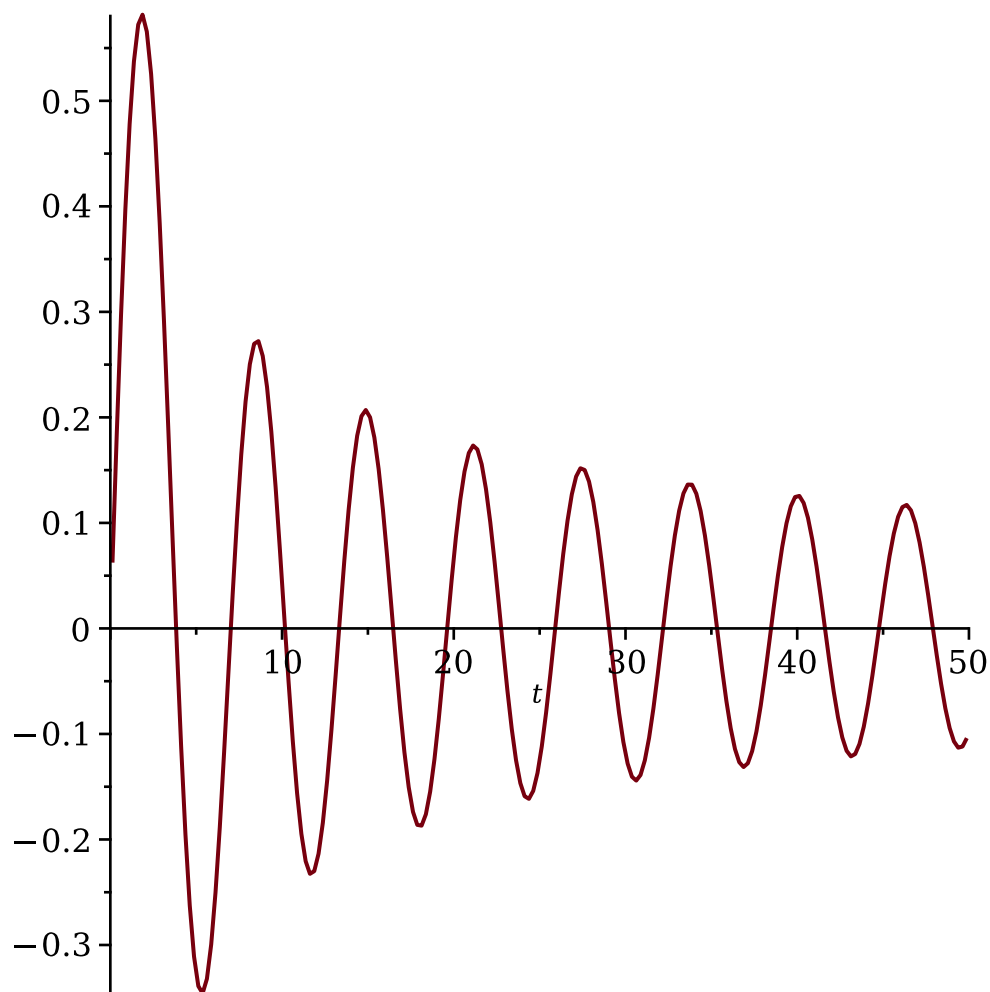
-> Bessel

<- Bessel successful

<- special function solution successful

$$x(t) = c_1 \sqrt{t} \text{BesselJ}\left(\frac{1}{7}, \frac{2t^{7/2}}{7}\right) + c_2 \sqrt{t} \text{BesselY}\left(\frac{1}{7}, \frac{2t^{7/2}}{7}\right) \quad (99)$$

```
> plot(BesselJ(1,t),t=0..50);
```

```
> restart;
```

```
> # Exercise 5
```

```
> # a)
```

```
> eqa:=diff(x(t),t$2)+5*x(t)=0;
```

$$eqa := \frac{d^2}{dt^2} x(t) + 5 x(t) = 0 \quad (100)$$

```
> ica:=x(0)=0,D(x)(0)=0;
```

$$ica := x(0) = 0, D(x)(0) = 0 \quad (101)$$

```
> dsolve({eqa,ica},x(t));
```

$$x(t) = 0 \quad (102)$$

```
> # b)
```

```
> eqb:=diff(x(t),t$2)+t*x(t)=0;
```

$$eqb := \frac{d^2}{dt^2} x(t) + t x(t) = 0 \quad (103)$$

```
> icb:=x(0)=0,D(x)(0)=0;
```

$$icb := x(0) = 0, D(x)(0) = 0 \quad (104)$$

```
> dsolve({eqb,icb},x(t));
```

$$x(t) = 0 \quad (105)$$

```

> # c)
> eqc:=diff(x(t),t$2)+t^5*x(t)=0;
                                
$$eqc := \frac{d^2}{dt^2} x(t) + t^5 x(t) = 0$$

                                (106)
=
> icc:=x(0)=0,D(x)(0)=0;
                                
$$icc := x(0) = 0, D(x)(0) = 0$$

                                (107)
=
> dsolve({eqc,icc},x(t));
                                
$$x(t) = c_1 \sqrt{t} \text{BesselJ}\left(\frac{1}{7}, \frac{2t^{7/2}}{7}\right)$$

                                (108)
=
> restart;
> # Exercise 6
> # a)
> eqa:=diff(x(t),t$2)+x(t)=0;
                                
$$eqa := \frac{d^2}{dt^2} x(t) + x(t) = 0$$

                                (109)
=
> ica:=x(0)=0,x(Pi)=0;
                                
$$ica := x(0) = 0, x(\pi) = 0$$

                                (110)
=
> dsolve({eqa,ica},x(t));
                                
$$x(t) = c_1 \sin(t)$$

                                (111)
=
> # b)
> icb:=x(0)=0,x(1)=0;
                                
$$icb := x(0) = 0, x(1) = 0$$

                                (112)
=
> dsolve({eqa,icb},x(t));
                                
$$x(t) = 0$$

                                (113)
=
> # c)
> eqc:=diff(x(t),t$2)+x(t)=1;
                                
$$eqc := \frac{d^2}{dt^2} x(t) + x(t) = 1$$

                                (114)
=
> icc:=x(0)=0,x(Pi)=0;
                                
$$icc := x(0) = 0, x(\pi) = 0$$

                                (115)
=
> dsolve({eqc,icc},x(t));
> restart;
> # Exercise 7
> # a)
> eqa:=diff(x(t),t)+x(t)=15;
                                
$$eqa := \frac{d}{dt} x(t) + x(t) = 15$$

                                (116)
=
> dsolve(eqa,x(t));
                                
$$x(t) = 15 + e^{-t} c_1$$

                                (117)
=
> # b)

```

```
> eqb:=diff(x(t),t)+x(t)=2*exp(t)-7*exp(-3*t);
```

$$eqb := \frac{d}{dt} x(t) + x(t) = 2e^t - 7e^{-3t} \quad (118)$$

```
> dsolve(eqb,x(t));
```

$$x(t) = \frac{7e^{-3t}}{2} + e^t + e^{-t} c_1 \quad (119)$$

```
> # c)
> eqc:=diff(x(t),t)+x(t)=-t^2+3*t-7;
```

$$eqc := \frac{d}{dt} x(t) + x(t) = -t^2 + 3t - 7 \quad (120)$$

```
> dsolve(eqc,x(t));
```

$$x(t) = -t^2 + 5t - 12 + e^{-t} c_1 \quad (121)$$

```
> # d)
> eqd:=diff(x(t),t)+x(t)=sin(t)+3*cos(t);
```

$$eqd := \frac{d}{dt} x(t) + x(t) = \sin(t) + 3\cos(t) \quad (122)$$

```
> dsolve(eqd,x(t));
```

$$x(t) = 2\sin(t) + \cos(t) + e^{-t} c_1 \quad (123)$$

```
> # e)
> eqe:=diff(x(t),t)+x(t)=sin(t);
```

$$eqe := \frac{d}{dt} x(t) + x(t) = \sin(t) \quad (124)$$

```
> dsolve(eqe,x(t));
```

$$x(t) = -\frac{\cos(t)}{2} + \frac{\sin(t)}{2} + e^{-t} c_1 \quad (125)$$

```
> # f)
> eqf:=diff(x(t),t)+x(t)=3*cos(t);
```

$$eqf := \frac{d}{dt} x(t) + x(t) = 3\cos(t) \quad (126)$$

```
> dsolve(eqf,x(t));
```

$$x(t) = \frac{3\cos(t)}{2} + \frac{3\sin(t)}{2} + e^{-t} c_1 \quad (127)$$